KAIZEN MATHS

Lesson starter

Product, Quotient and Chain Rule

Trig and Log Differentiation

Name _____

S1

Differentiate with respect to x.

$y = 3x^2 \cos(4x)$

S2

Differentiate with respect to x.

$y = (3x^4 + 3)\ln(x)$

S3

Differentiate with respect to x.

$y = 2x^2 \sin(3x)$

S4

Differentiate with respect to x.

$y = (5x^3 + 3)\ln(x)$

Show your working

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ALG-1258-L3 · S

Show your working

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Exit ticket

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ALG-1258-L3 · E

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ALG-1258-L3 · E

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Show your working

Teacher answers & working

Product, Quotient and Chain Rule · Trig and Log Differentiation

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S1

Differentiate with respect to x.

$y = 3x^2 \cos(4x)$

Answer:

$\frac{dy}{dx} = 6x\cos(4x) - 12x^2\sin(4x)$

Working:

$u = 3x^2, \quad v = \cos(4x)$

$u' = 3 \cdot 2x^{2-1}$

$u' = 6x$

$w = 4x, \quad w' = 4$

$v' = -\sin(w)w'$

$v' = -\sin(4x) \cdot 4$

$y' = u'v + uv'$

$y' = (6x)(\cos(4x)) + (3x^2)[-\sin(4x) \cdot 4]$

$\frac{dy}{dx} = 6x\cos(4x) - 12x^2\sin(4x)$

S2

Differentiate with respect to x.

$y = (3x^4 + 3)\ln(x)$

Answer:

$\frac{dy}{dx} = 12x^3\ln(x) + \frac{3x^4 + 3}{x}$

Working:

$u = 3x^4 + 3, \quad v = \ln(x)$

$u' = 3 \cdot 4x^{4-1} + 0$

$u' = 12x^3$

$v' = \frac{1}{x}$

$y' = u'v + uv'$

$y' = (12x^3)\ln(x) + (3x^4 + 3) \cdot \frac{1}{x}$

$\frac{dy}{dx} = 12x^3\ln(x) + \frac{3x^4 + 3}{x}$

Teacher answers & working

Product, Quotient and Chain Rule · Trig and Log Differentiation

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S3

Differentiate with respect to x.

$y = 2x^2 \sin(3x)$

Answer:

$\frac{dy}{dx} = 4x \sin(3x) + 6x^2 \cos(3x)$

Working:

$u = 2x^2, \quad v = \sin(3x)$

$u' = 2 \cdot 2x^{2-1}$

$u' = 4x$

$w = 3x, \quad w' = 3$

$v' = \cos(w)w'$

$v' = \cos(3x) \cdot 3$

$y' = u'v + uv'$

$y' = (4x)(\sin(3x)) + (2x^2)[\cos(3x) \cdot 3]$

$\frac{dy}{dx} = 4x \sin(3x) + 6x^2 \cos(3x)$

S4

Differentiate with respect to x.

$y = (5x^3 + 3)\ln(x)$

Answer:

$\frac{dy}{dx} = 15x^2 \ln(x) + \frac{5x^3 + 3}{x}$

Working:

$u = 5x^3 + 3, \quad v = \ln(x)$

$u' = 5 \cdot 3x^{3-1} + 0$

$u' = 15x^2$

$v' = \frac{1}{x}$

$y' = u'v + uv'$

$y' = (15x^2)\ln(x) + (5x^3 + 3) \cdot \frac{1}{x}$

$\frac{dy}{dx} = 15x^2 \ln(x) + \frac{5x^3 + 3}{x}$

Teacher answers & working

Product, Quotient and Chain Rule · Trig and Log Differentiation

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E1

Differentiate with respect to x.

$y = 4x^2 \sin(2x)$

Answer:

$\frac{dy}{dx} = 8x \sin(2x) + 8x^2 \cos(2x)$

Working:

$u = 4x^2, \quad v = \sin(2x)$

$u' = 4 \cdot 2x^{2-1}$

$u' = 8x$

$w = 2x, \quad w' = 2$

$v' = \cos(w)w'$

$v' = \cos(2x) \cdot 2$

$y' = u'v + uv'$

$y' = (8x)(\sin(2x)) + (4x^2)[\cos(2x) \cdot 2]$

$\frac{dy}{dx} = 8x \sin(2x) + 8x^2 \cos(2x)$

E2

Differentiate with respect to x.

$y = (2x^3 + 3)\ln(x)$

Answer:

$\frac{dy}{dx} = 6x^2 \ln(x) + \frac{2x^3 + 3}{x}$

Working:

$u = 2x^3 + 3, \quad v = \ln(x)$

$u' = 2 \cdot 3x^{3-1} + 0$

$u' = 6x^2$

$v' = \frac{1}{x}$

$y' = u'v + uv'$

$y' = (6x^2)\ln(x) + (2x^3 + 3) \cdot \frac{1}{x}$

$\frac{dy}{dx} = 6x^2 \ln(x) + \frac{2x^3 + 3}{x}$

Teacher answers & working

Product, Quotient and Chain Rule · Trig and Log Differentiation

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E3

Differentiate with respect to x.

$y = 2x^2 \cos(4x)$

Answer:

$\frac{dy}{dx} = 4x\cos(4x) - 8x^2\sin(4x)$

Working:

$u = 2x^2, \quad v = \cos(4x)$

$u' = 2 \cdot 2x^{2-1}$

$u' = 4x$

$w = 4x, \quad w' = 4$

$v' = -\sin(w)w'$

$v' = -\sin(4x) \cdot 4$

$y' = u'v + uv'$

$y' = (4x)(\cos(4x)) + (2x^2)[-\sin(4x) \cdot 4]$

$\frac{dy}{dx} = 4x\cos(4x) - 8x^2\sin(4x)$